AVL TREE SIMULATION AND TRAVERSAL

Tool-VisuAlgo

CO3-AT2

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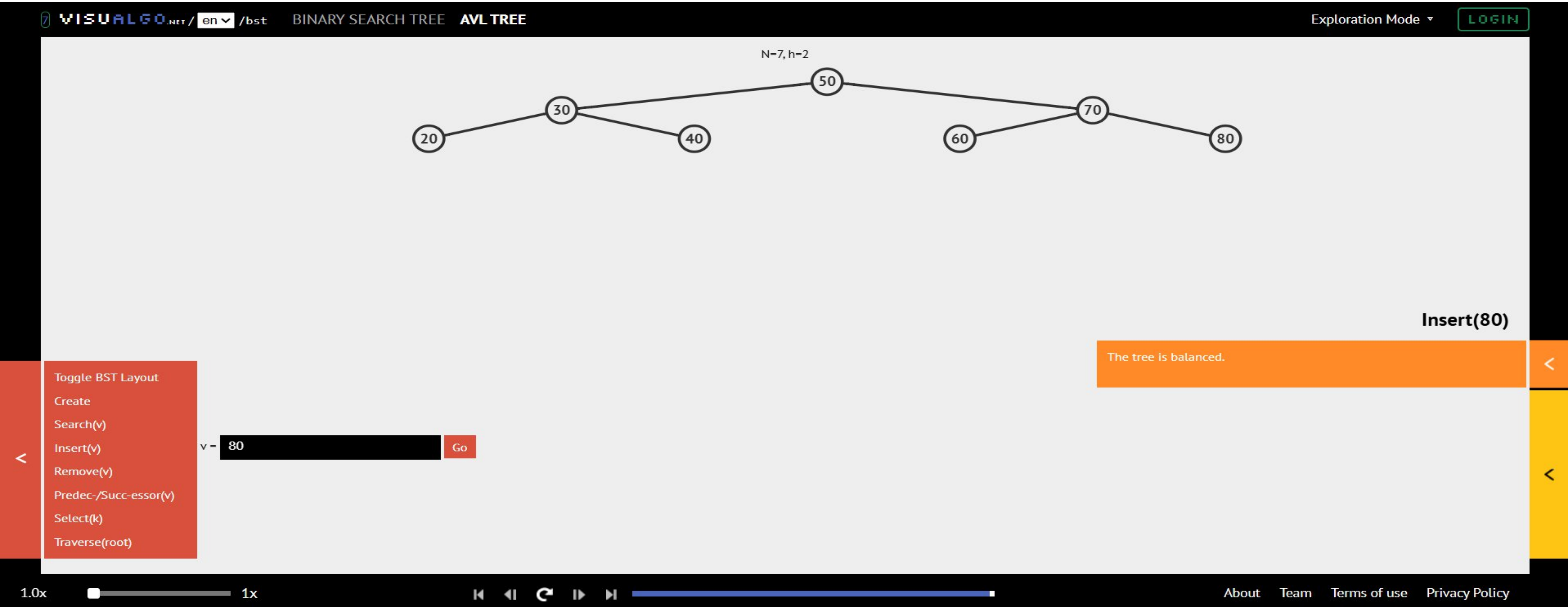
OPERATIONS:

1. INSERTION
2. DELETION
3. SEARCHING
4. TRAVERSING
   i. IN-ORDER
   ii. PRE-ORDER
   iii. POST-ORDER
5. ROTATION
   i. LEFT-SIDE
   ii. RIGHT-SIDE

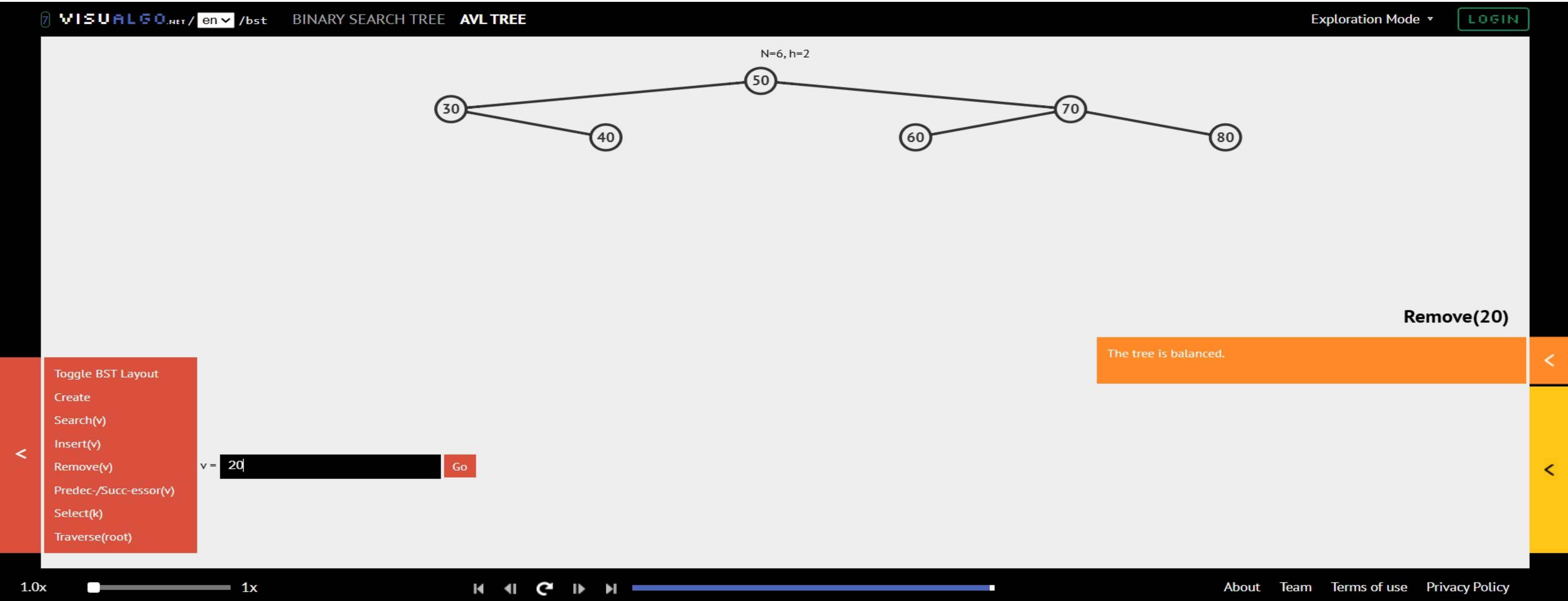
1.INSERTION:

- Select Insert in the AVL simulator and enter 50, 30, 70, 20, 40, 60, 80 one by one.
- The simulator places each value according to the BST property while maintaining AVL balance.
- The final tree contains 3 levels with 50 as the root.
- No rotation is required because the tree remains balanced.



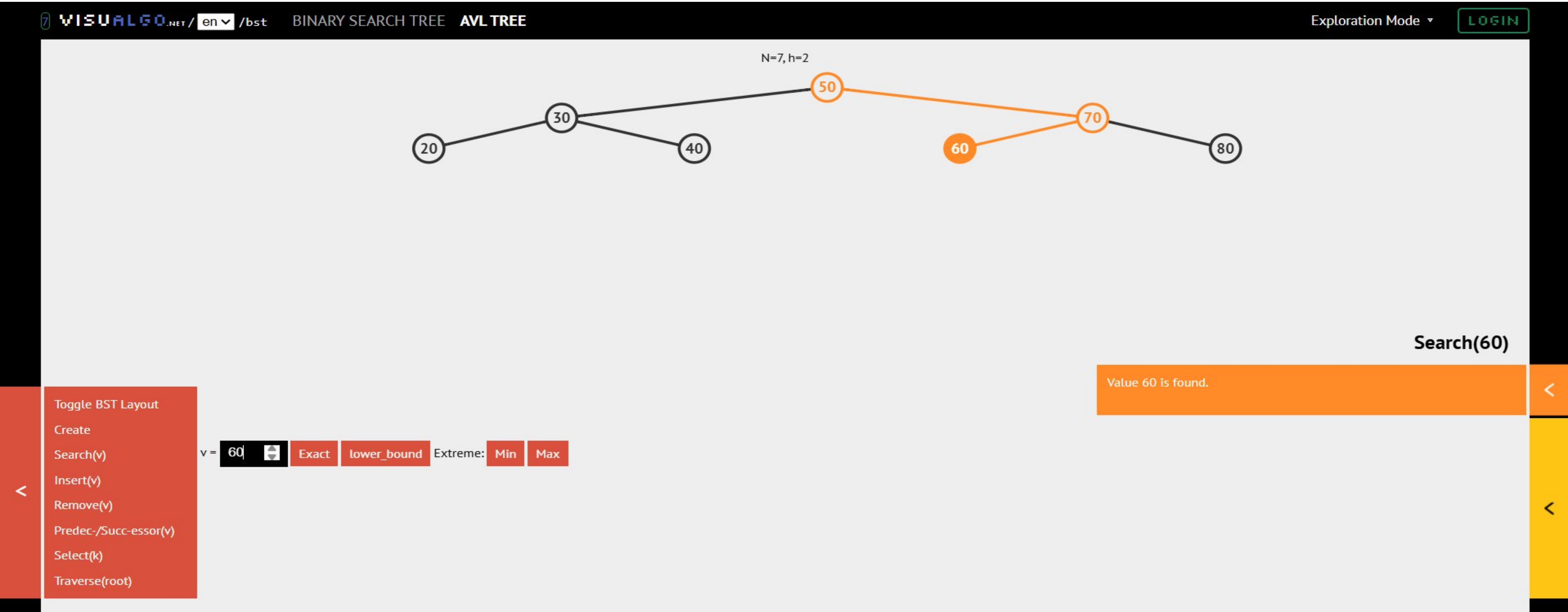
2. DELETION:

- Select Delete and enter 20.
- The simulator follows 50 → 30 → 20 and removes node 20.
- Since 20 is a leaf node, it can be directly deleted.
- The remaining AVL tree stays balanced.



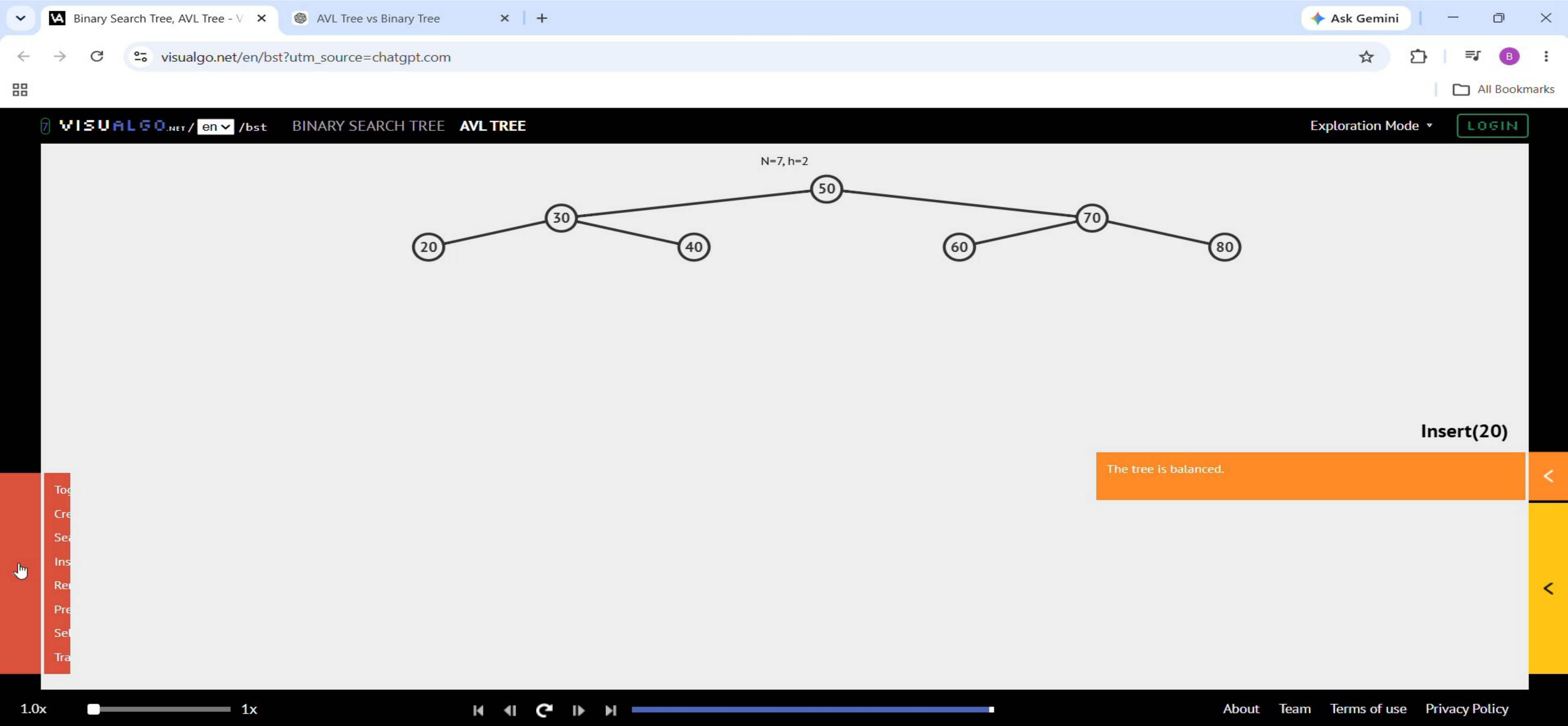
3. SEARCHING:

- Select Search and enter 60.
- The simulator compares 60 with 50 and moves right to 70.
- Since 60 < 70, it moves left and reaches 60.
- Search path: 50 → 70 → 60 — 60 found.

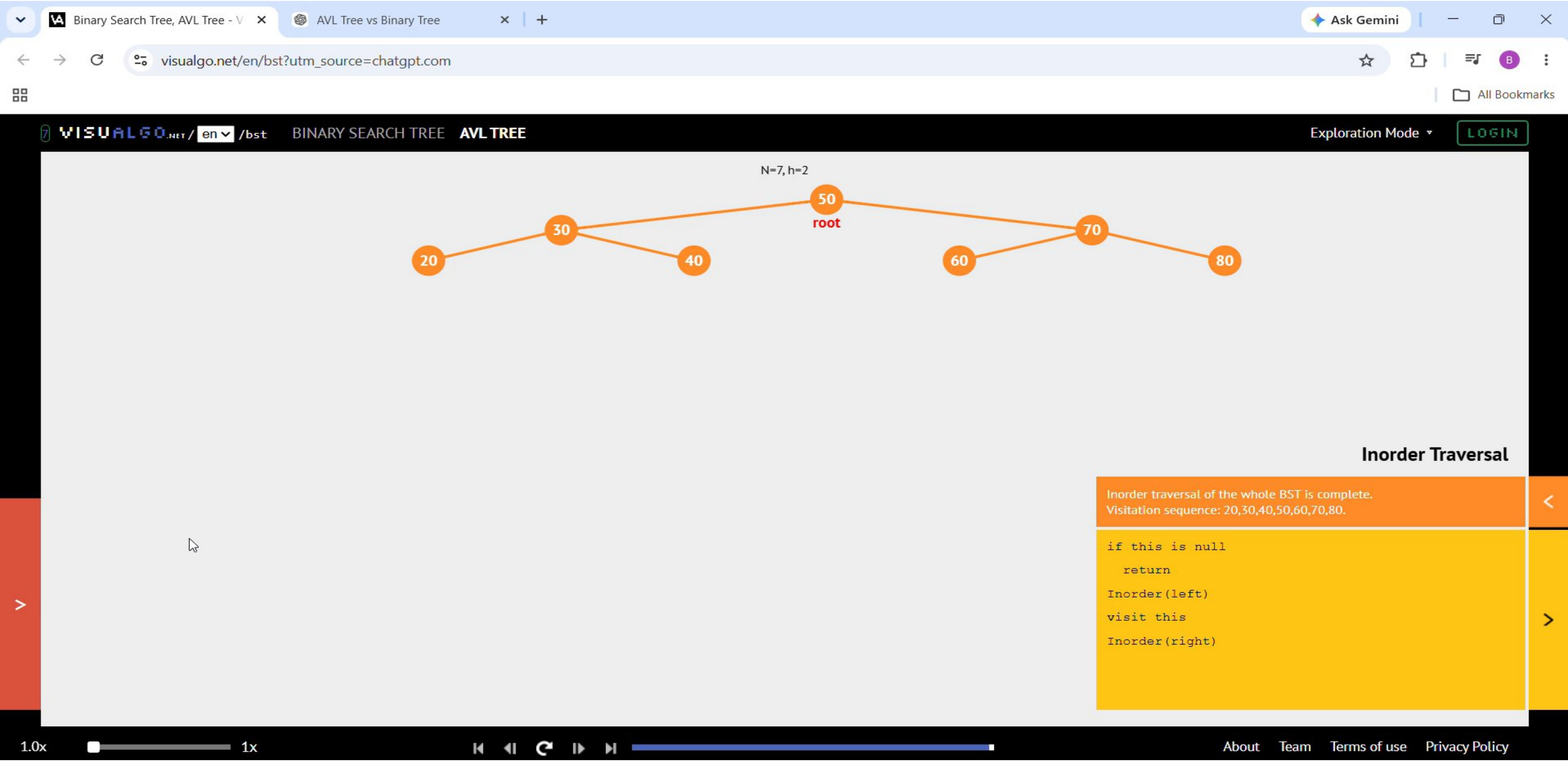


4. TRAVERSING:

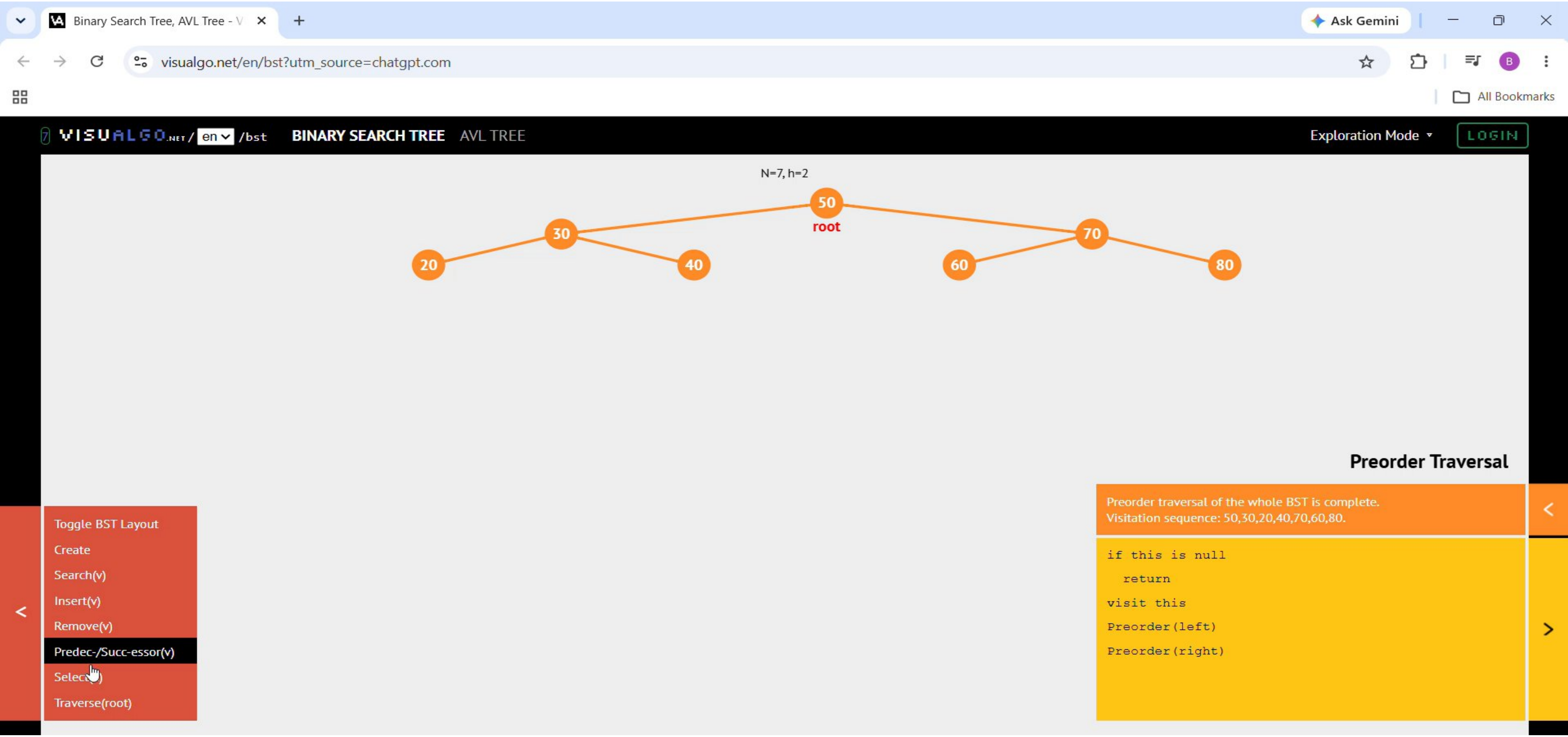
i. IN-ORDER: (Left – Root – Right)



ii. PRE-ORDER: (Root – Left – Right)

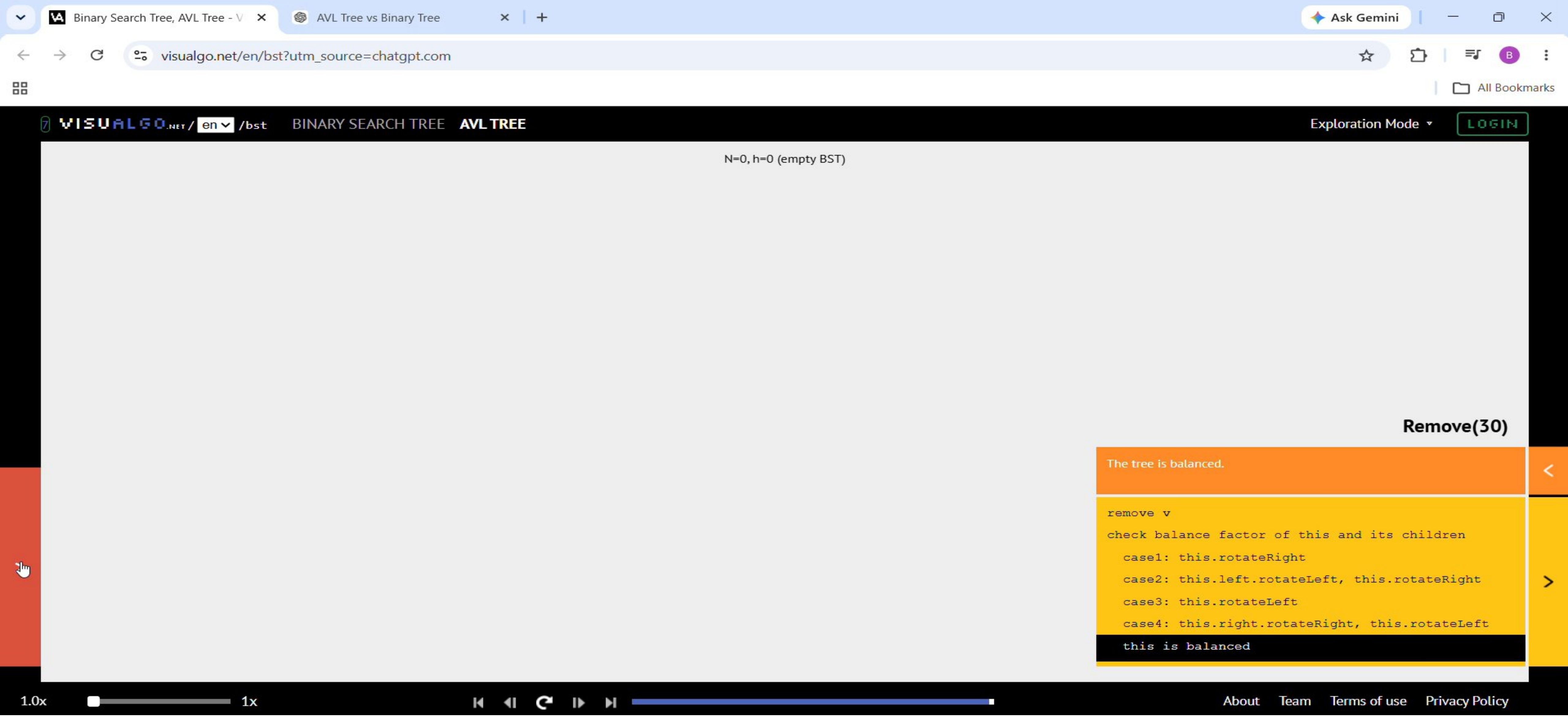


iii. POST-ORDER: (Left – Right – Root)

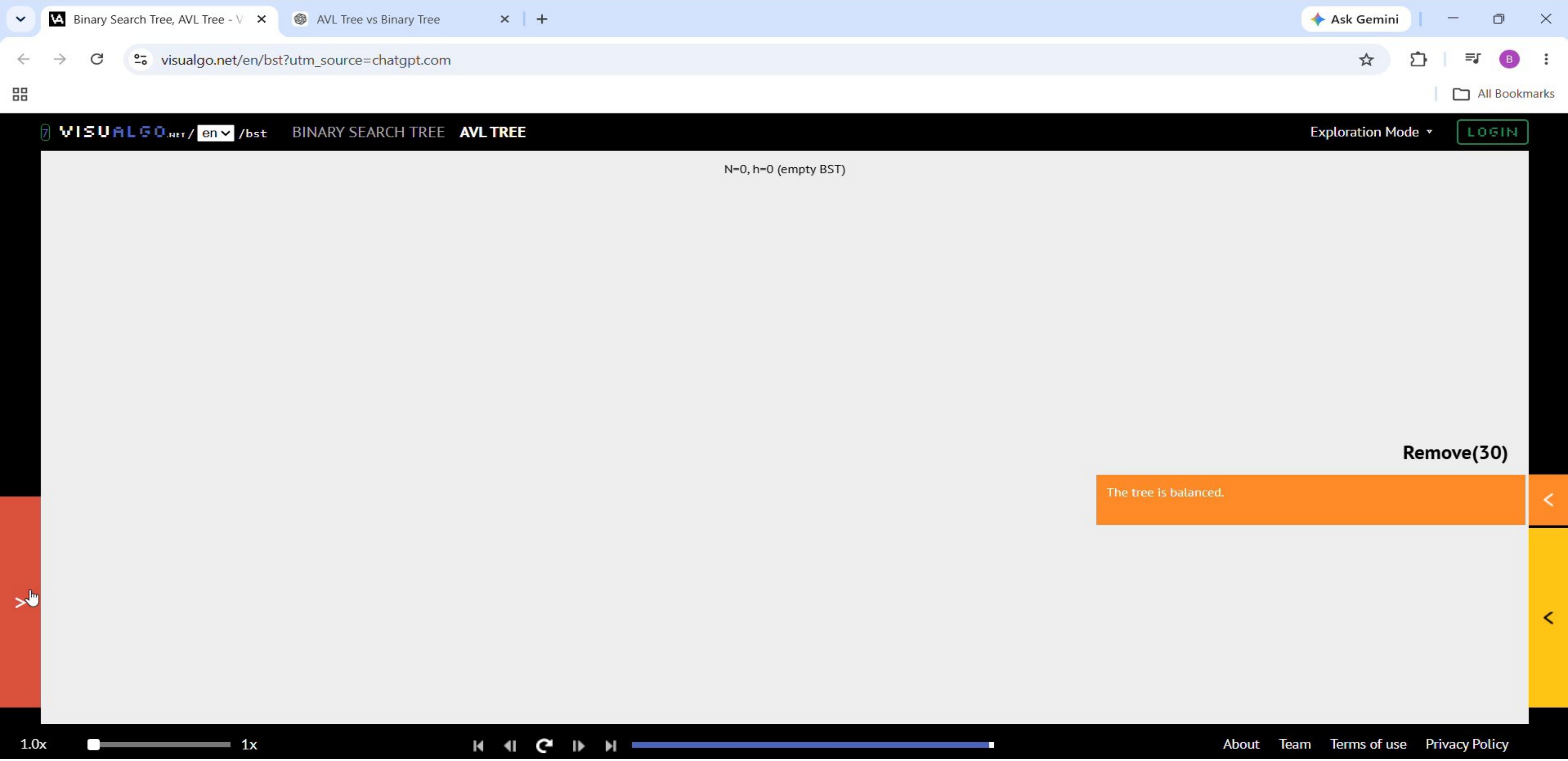


5. ROTATION:

i. LEFT-SIDE:



ii. RIGHT-SIDE:



Thank you